Cynbit website UX consistency & navigation enhancement

Technical report · Full Stack Developer Intern assessment, Cynbit Technologies · 31 October 2026

1. Assessment overview

The brief: analyse https://www.cynbit.in from a user and a developer perspective, pick one meaningful problem, and deliver a working prototype that **improves the existing site**. This submission keeps Cynbit's branding and presents each change as a before/after on the same pages, rather than a new site in a similar style. It is an independent prototype; the production site was not modified.

2. Problem statement

Three related inconsistencies make the site harder to scan and to navigate: (1) the Learning Paths cards vary in size, shape and content; (2) the Our Story timeline cards are visually weak and the year is nearly invisible; (3) nothing tells a visitor which page or section they are in. They share one cause, an absence of a consistent visual and orientation system, so they are treated as one project: **Website UX Consistency & Navigation Enhancement**.

3. Existing website analysis

How the site was inspected, and what could not be

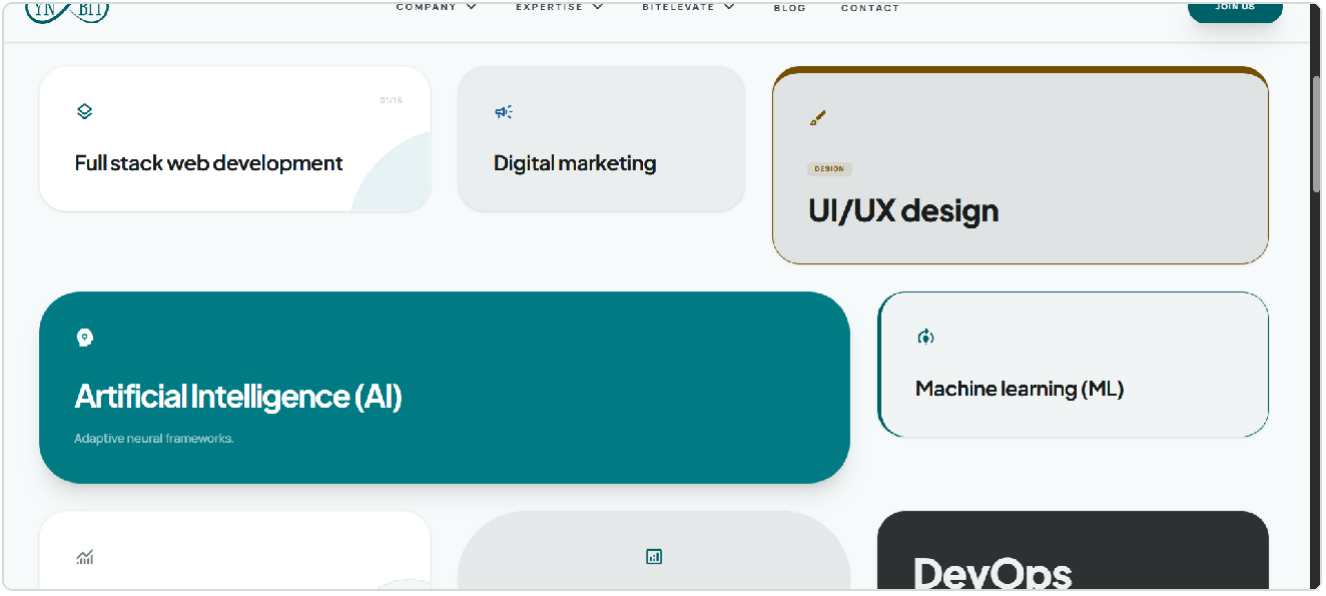
The build environment could not open cynbit.in in a browser and its shell was blocked from the site, so the evidence comes from three sources: (a) page text and structure read with a web-fetch tool; (b) screenshots supplied by the candidate, including Chrome Lighthouse results (desktop and mobile), the DevTools Elements panel on /contact, and the DevTools Network panel on /team; (c) the Next.js RSC payload of /story, supplied as a text file. **The Lighthouse and DevTools runs were made by the candidate on the live site; I analysed their output but did not run them.** Pages read: Home, Our Story, Team, Learning Paths, Careers, Impact, Services, Products, Case Studies, BitElevate, Student Diaries, Blog, Contact, Navigation.

Area	Observed (source)
Navigation	Header: Company (Our Story, Our Team, Careers, Social Impact), Expertise (Services Hub, Product Suite, Case Studies), BitElevate (The Ecosystem, Learning Paths, Student Diaries), Blog, Contact, Join Us. Footer columns, legal links and a five-item mobile bottom bar (Home, Services, Products, BitElevate, Contact). (fetched pages)
Active state / breadcrumbs	The mobile bottom bar fills a teal pill on the current item for exactly the five top-level pages (Home on the home page, Contact on /contact; screenshots). Sub-pages such as Our Story, Team and Learning Paths highlight nothing, and no breadcrumb appears

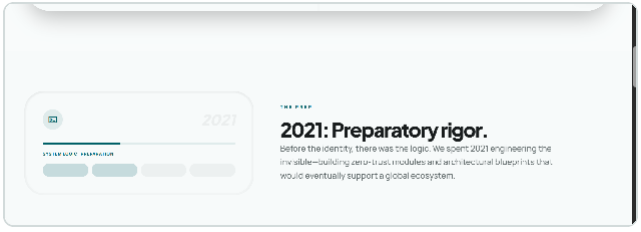
	in the page content retrieved. The desktop header's active state was not captured in any screenshot and remains unverified.
Learning Paths	16 cards. The supplied screenshot shows mixed widths (about 567, 416 and 718 px in one row), heights (about 210, 277 and 285 px), surfaces (white, grey, brown-topped, teal, outlined, charcoal) and radii; the counter "01/16" and the DESIGN tag appear on one card each; most cards have no description.
Our Story	Left card: faint italic year in near-white grey, thin progress bar with no labels, four unlabelled pills (two filled, two empty). (screenshot)
Technology	Next.js App Router with a `(marketing)` route group (RSC payload of /story, `_rsc=` prefetches, `/_next/static/chunks`); fonts through `next/font` (`__variable_*` classes, self-hosted woff2); Material Symbols Outlined loaded from Google Fonts; Tailwind-style utility classes (`fixed top-0 z-[100]`, `bg-surface`, `@layer utilities`); Google Analytics (`G-LK6CQWCWX0`); JSON-LD `ProfessionalService`. Content is server-rendered/static; no API integration observed.

4. User-perspective observations

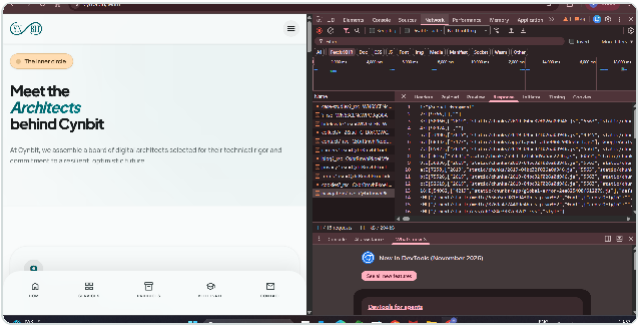
- **Expertise cards:** uneven dimensions and spacing; unbalanced layout; inconsistent content (counter, tag and description on a few cards only).
- **Our Story cards:** low-contrast year; unlabelled progress bar and pills carry no information; card looks empty next to a strong heading.
- **Navigation:** the bottom bar marks only the five top-level pages, so Our Story, Team and Learning Paths show no current-page state; dropdown selection gives no lasting feedback; no breadcrumb.



Original: expertise cards (screenshot supplied with the assessment)



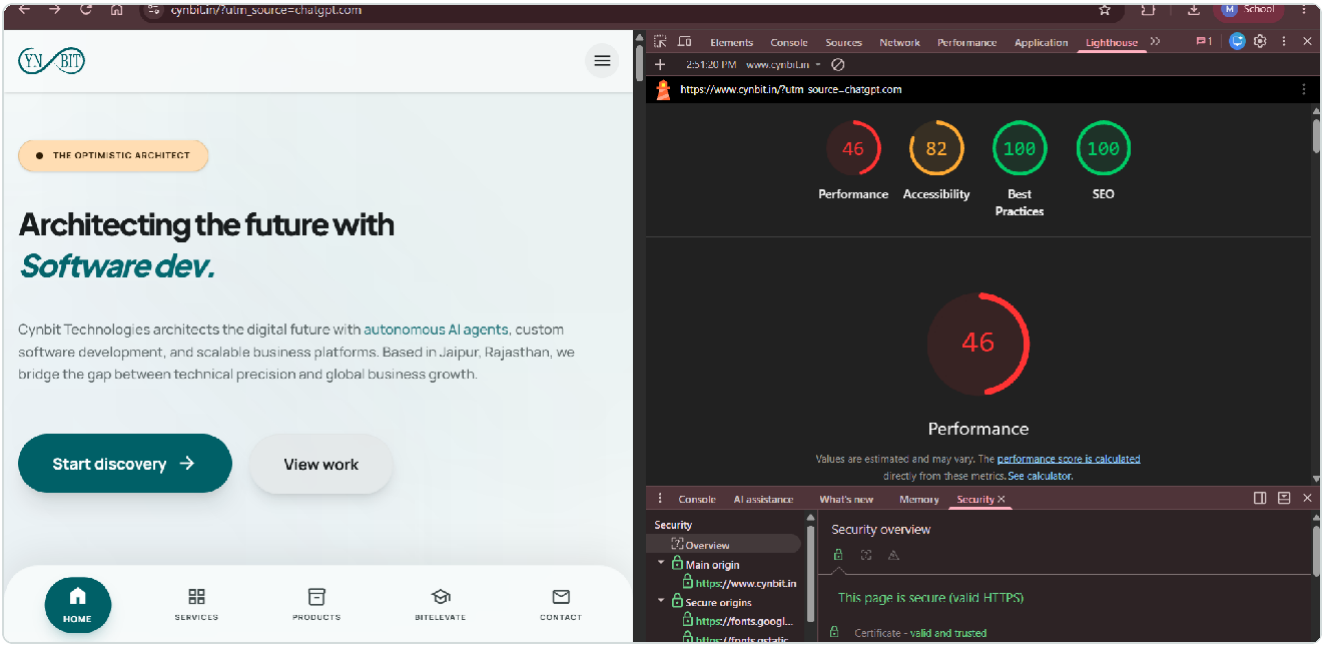
Original: Our Story timeline card (screenshot supplied)



Original: Team page on mobile with DevTools network panel (screenshot supplied)

5. Developer-perspective findings

Evidence is the candidate's own Lighthouse and DevTools output from the live site (see section 3). Statements marked *inference* are my reading of that evidence, not something measured.



Live site: Lighthouse on the home page (Performance 46, Accessibility 82, Best Practices 100, SEO 100), supplied by the candidate

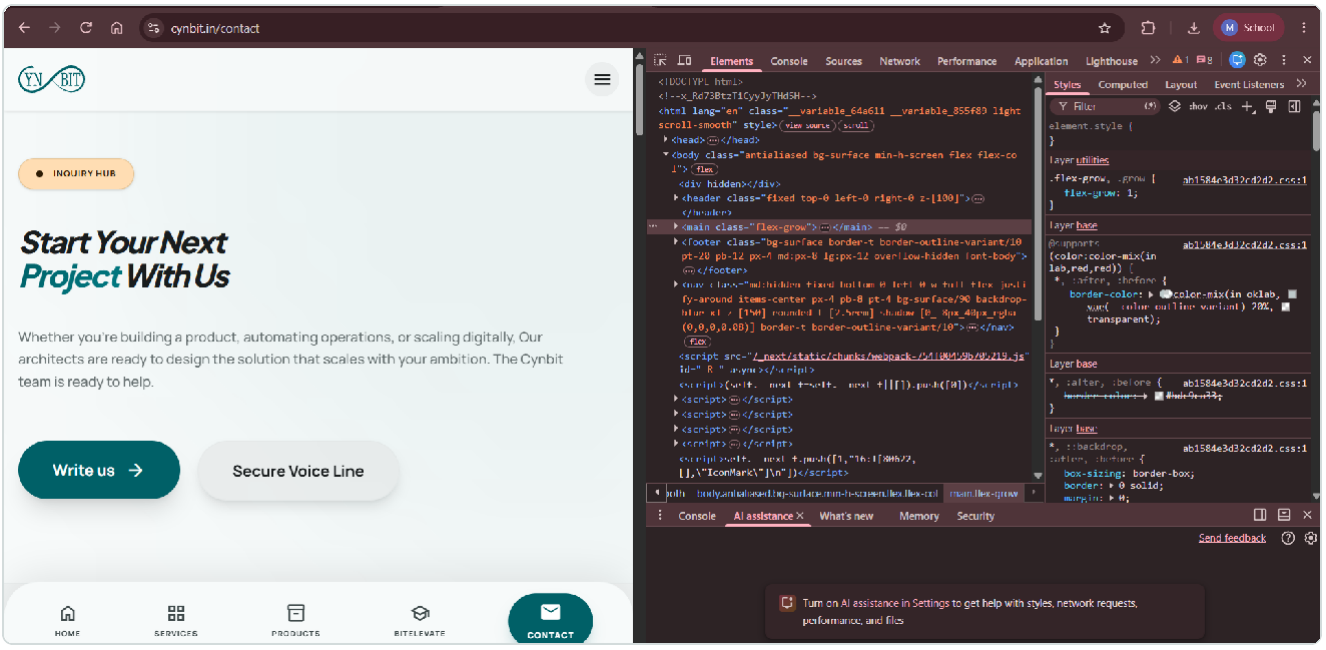
5.1 Lighthouse results (live site)

	Performance	Accessibility	Best practices	SEO	FCP	LCP	TBT	CLS	Speed Index
Desktop	46	82	100	100	2.0 s	2.1 s	2,690 ms	0	3.1 s
Mobile	not in the data supplied	82	77	100	9.1 s	10.8 s	0 ms	0	18.8 s

5.2 Findings

#	Finding	Evidence	Likely cause / impact
1	**Heavy main-thread JavaScript.** TBT 2,690 ms on desktop; 15 long tasks (desktop) and 12 (mobile); main-thread work 6.0 s / 7.7 s; unused JavaScript 158 KiB / 122 KiB; legacy JavaScript 11 KiB; forced reflow flagged on desktop.	Lighthouse	*Inference:* a large client bundle. The /story payload shows the header, footer and bottom navigation shipped as client components in the shared layout, plus a client Google Analytics component in the root layout.
2	**Render-blocking requests.** Estimated savings 1,520 ms desktop and 5,760 ms mobile. Mobile FCP 9.1 s, LCP 10.8 s, Speed Index 18.8 s. The LCP element is a hero text paragraph, not an image.	Lighthouse; /story payload (`<head>` contains a stylesheet link to fonts.googleapis.com for Material Symbols Outlined, plus two CSS files)	*Inference:* the icon-font stylesheet and CSS delay first paint. The per-request detail list was not supplied.
3	**Accessibility 82 on both runs.** Insufficient colour contrast; lists with non-li children and li outside lists; heading levels out of order; touch targets too small or too close (desktop); links distinguished by colour alone (mobile).	Lighthouse	The specific elements were not in the data supplied. Each is a WCAG-level issue affecting screen-reader and touch users.
4	**Mobile best practices 77:** third-party cookies (30 found) and issues logged in the Chrome Issues panel. Manual-check hints on both runs: CSP, HSTS, COOP, X-Frame-Options/CSP, Trusted Types.	Lighthouse	Third-party cookies point to Google Analytics or embedded services (*inference*). Security headers are suggestions, not failures.
5	**Page metadata on /story.** `og:url` is `https://cynbit.in` (the site root) instead of the page address; `og:image` and `twitter:image` point to `/logo.svg`; the title repeats the brand ("... | Cynbit Technologies | Cynbit Technologies"). The JSON-LD geo coordinates read 26.8000 / 75.8000.	/story RSC payload	SEO scores 100 because Lighthouse does not check these. A wrong `og:url` can make shared links resolve to the home page; many platforms do not render SVG previews. The coordinates look rounded (*inference*).
6	**Navigation is implemented once, in a shared layout.** Header, footer and bottom navigation are client components of the `(marketing)` layout, shared by every page; each page also gets a template. Header is	/story payload; Elements panel on /contact	This is why the improvement is cheap in the real app: one active-state rule in the shared header and bottom bar, driven by

	`fixed top-0 z-[100]`; the bottom nav is `md:hidden fixed bottom-0 z-[150]`.		`usePathname()`, covers every page.
7	**Console / issue indicators.** The Elements screenshot on /contact shows a warning badge (1) and a second badge (8) in the DevTools toolbar.	Elements screenshot	The messages themselves were not supplied, so no diagnosis is made.



Live site: Elements panel on /contact showing the fixed header, fixed bottom nav and Tailwind-style classes; the bottom bar shows Contact selected. Supplied by the candidate

5.3 Inspected

Console message text and failed network requests; form validation and API behaviour on Contact, Products and BitElevate; status codes of every link; the desktop header's active state; the elements behind each Lighthouse accessibility failure; keyboard and screen-reader behaviour.

5.4 What the prototype does about these findings

Live finding	Prototype status
Heading order	My own check found the same defect in the first prototype build (card titles were h3 under the page h1 on Learning Paths). Fixed (cards are h2); now tested on 5 pages at 2 widths.
List markup	Every list is ul/ol with li children only; tested.
Touch targets	The first build had footer links 18 to 21 px tall. Fixed to 44 px; all controls tested at 24 px or more (inline links in sentences are exempt under WCAG 2.5.8).
Contrast	Measured for the token pairs the prototype uses: body ink on page 16.4:1, secondary text 7.3:1 and 5.3:1, text on grey cards 5.0:1, teal on white 7.3:1, white on teal 5.1:1, white on charcoal 13.2:1. All at least 4.5:1. This is not a full-page audit.

Active navigation	Implemented for header, dropdown, mobile menu, bottom bar and footer, plus a breadcrumb. Tested.
Render-blocking and JS weight	**Not improved.** The prototype has no framework or analytics: 33 KB CSS (7.1 KB gzip) and 5.2 KB JS (1.8 KB gzip), a 20 KB page. These are payload sizes only. No Lighthouse score was run on the prototype, so no performance claim is made. It still loads one render-blocking Google Fonts stylesheet.
Metadata (og:url, SVG share image)	Not addressed; the prototype is marked noindex.

6. Selected improvement and justification

One project, three coordinated changes: consistent Learning Path cards, a stronger Our Story timeline, and persistent navigation orientation. They address the observations the user reported, sit on adjacent pages, and reuse one token set, so the result is coherent rather than three unrelated tweaks. The rest of the site was deliberately left alone.

Change	Design decision
Cards	Identical width and height per breakpoint (grid rows sized to 1fr), same anatomy on every card (icon chip, index, category, title, description), 4 / 2 / 1 columns so 16 cards form complete rows. The original surface variety is kept as colour only (white, grey, plus the teal and charcoal accents the site already uses), never size. Two-line title and description areas are reserved so tags line up across a row.
Timeline	Year set at 3.4 to 5.2 rem in brand teal on white (contrast about 7:1 against the page background (test requires at least 4.5:1), versus about 1.1:1 in the reconstruction of the original). Filled icon tile, labelled four-step rail that links to each year, chips or stats in place of empty pills, teal top accent, and a spine that fills with scroll.
Navigation	`aria-current="page"` is rendered from the URL for the item, its dropdown group, the mobile menu, bottom bar and footer; the active trigger gets colour, tint and an underline (not colour alone). A breadcrumb (Home, group, page) lives inside the sticky header, so state stays visible while scrolling. Dropdowns still close on selection, because staying open would obstruct the page; instead the parent remains highlighted and the current item is marked when reopened. Esc, outside click and focus-out close menus.

7. Technology stack

Dependency-free Node (ESM) static build, vanilla JavaScript for dropdowns, timeline and the view switch, plain CSS with custom properties, Playwright for tests. The live site uses Next.js; Next.js could not be installed in the build environment (npm registry blocked), so it was not used. See the README for the porting path.

8. Architecture and implementation

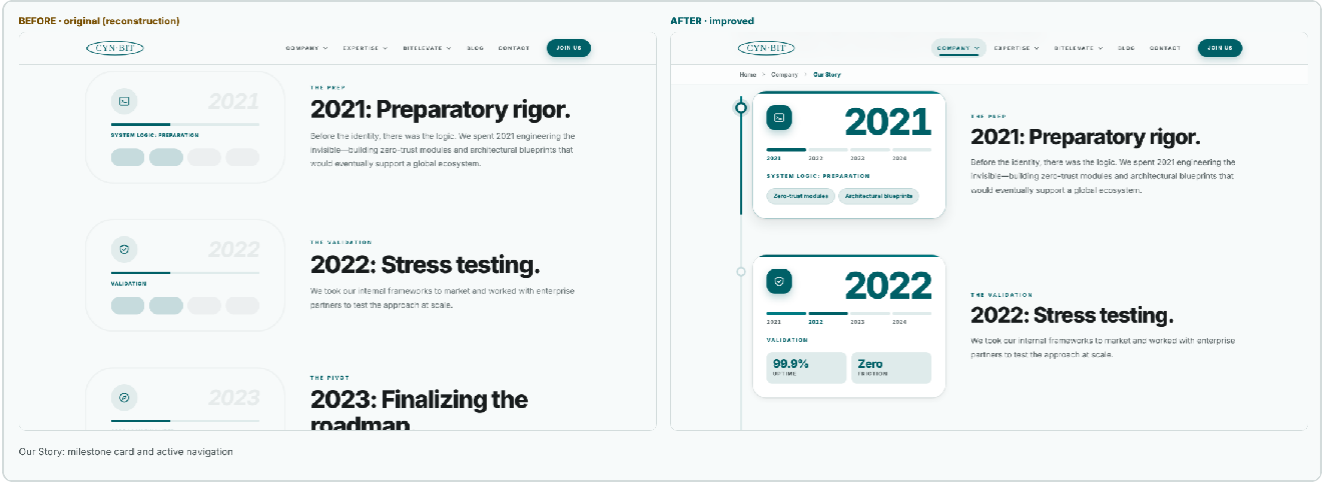
- `src/data/site.mjs` is the single source of truth for navigation. Header, breadcrumb, footer and bottom bar are all derived from it and from `findActiveRoute()` (longest-prefix match, so `/bitelevate/paths` marks Learning Paths, not The Ecosystem).
- Components are pure functions that return HTML; the active state is rendered at build time, so it works without JavaScript.
- `scripts/build.mjs` renders every route to `dist/<route>/index.html` and bundles CSS and JS. 18 pages are built; unbuilt site pages are route shells inside the same header and breadcrumb system with a link to the live page.
- The "original" view is a CSS skin scoped to `html[data-view='original']` (src/styles/original-skin.css), a reconstruction from screenshots, with no Cynbit code (including the live bottom bar, which marks only the five exact top-level pages). `/compare` shows both in two frames.
- Responsive design from 390 px to 1440 px, visible focus rings, skip link, labelled landmarks, reduced-motion support.

9. Screenshots: original website

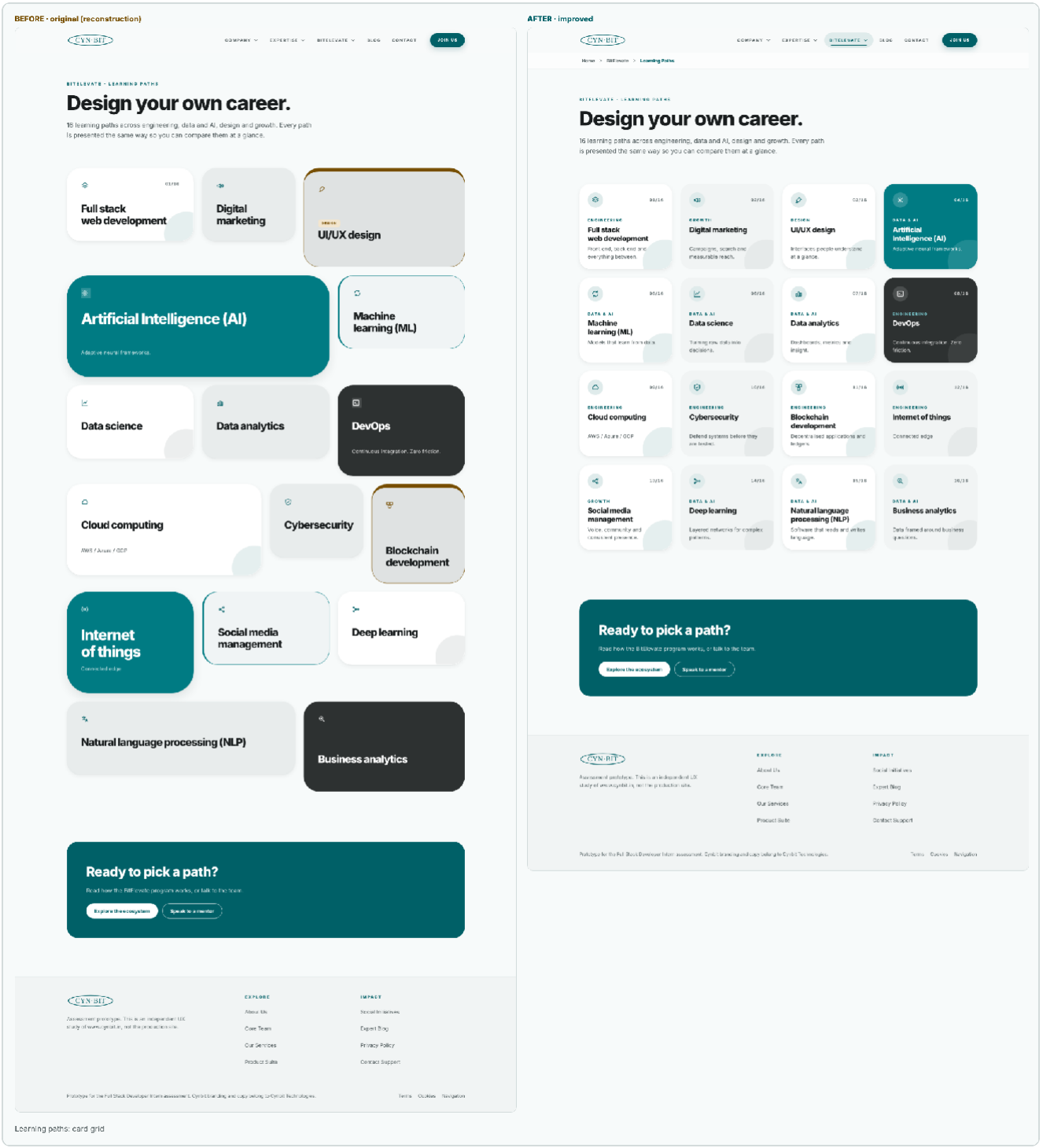
The three screenshots supplied with the assessment appear in section 4 and in `screenshots/original-reference/`. No other screenshots of the live site could be taken. Reconstruction renders are in `screenshots/reconstruction-original/`.

10. Screenshots: improved prototype and before/after

Left is the reconstruction of the original; right is the improved prototype. Fonts: the build environment could not load Google Fonts, so captures use the Inter fallback; with network access the site renders in Plus Jakarta Sans.

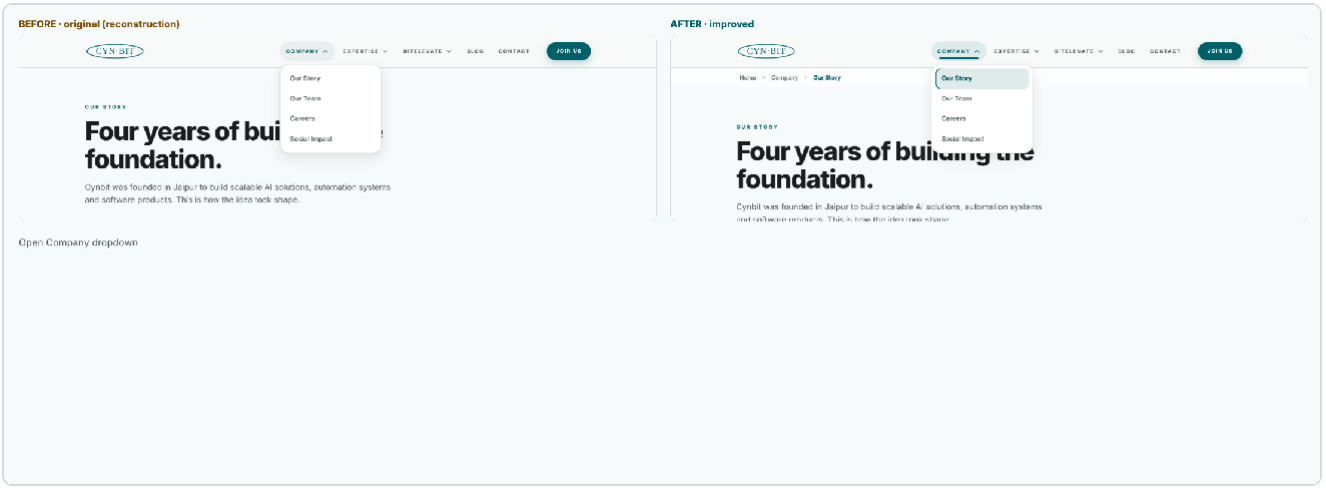


Our Story, desktop

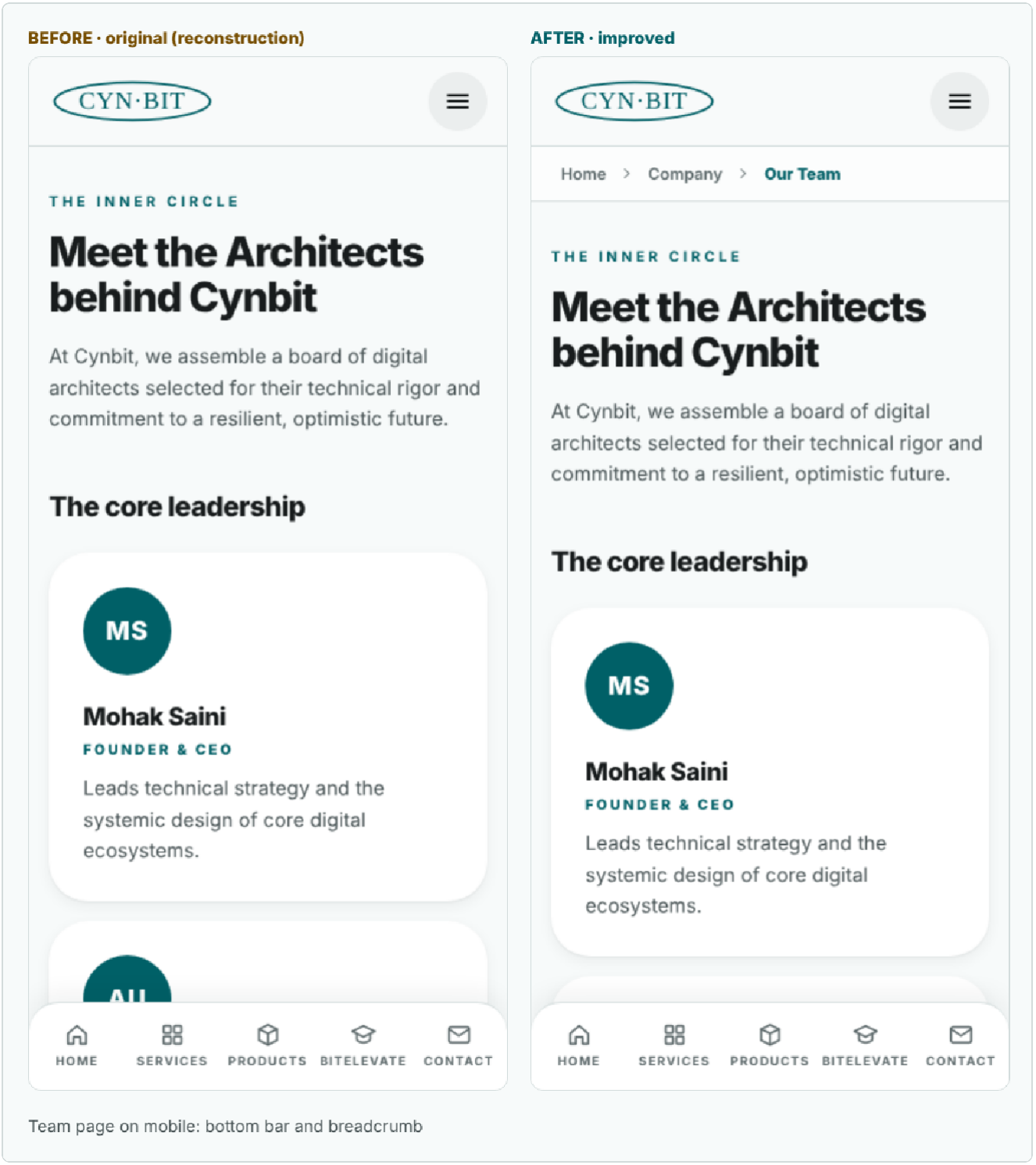


Learning paths: card grid

Learning paths, desktop



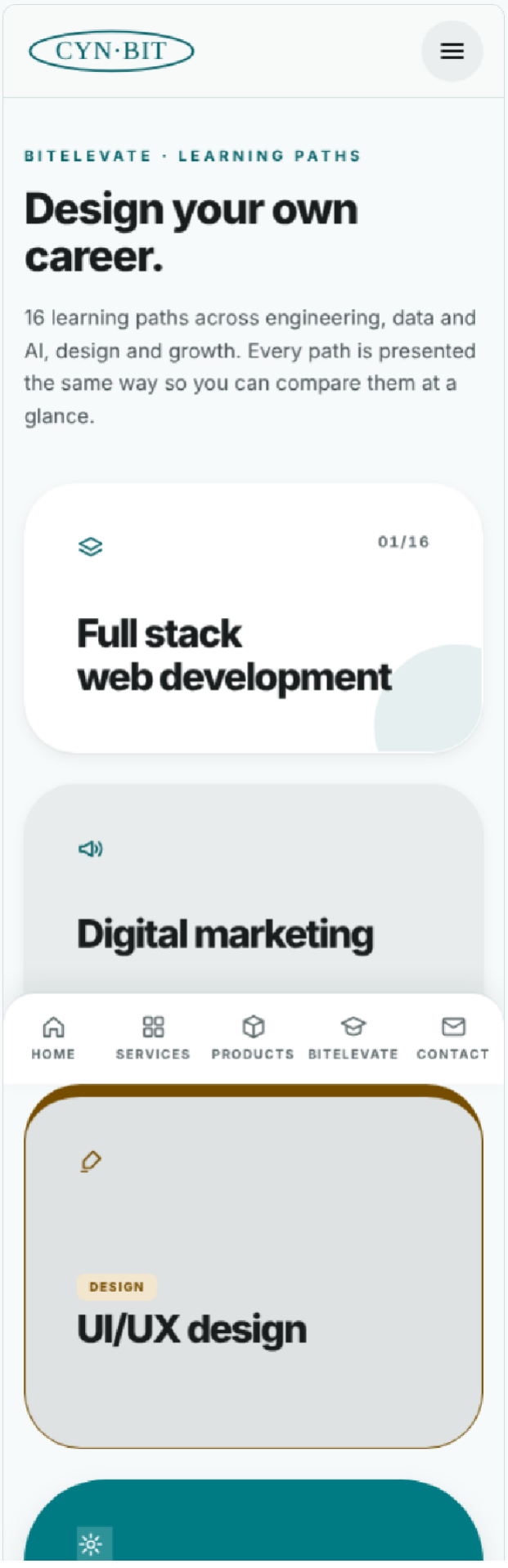
Navigation: breadcrumb, highlighted group, current item marked in dropdown



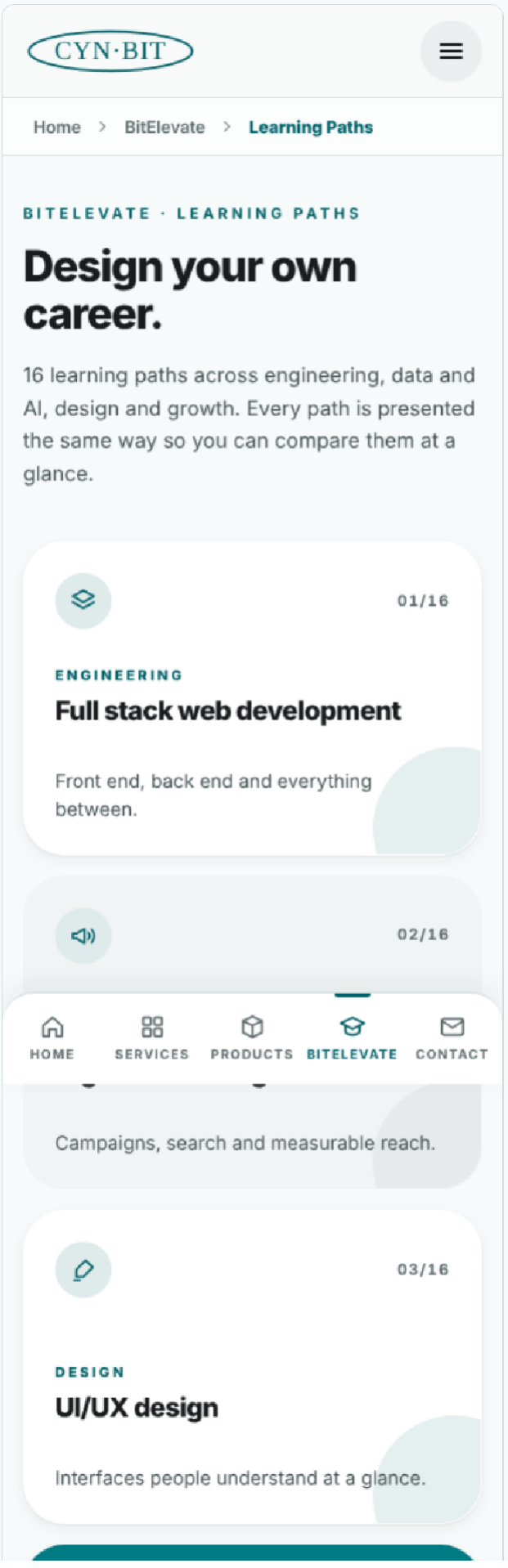
Team page on mobile: bottom bar and breadcrumb

Team page, mobile: breadcrumb under the sticky header

BEFORE · original (reconstruction)



AFTER · improved



Artificial Intelligence (AI)

Adaptive neural frameworks.

Machine learning (ML)

Data science

Data analytics

DevOps

Continuous integration. Zero friction.

04/16

DATA & AI

Artificial Intelligence (AI)

Adaptive neural frameworks.

05/16

DATA & AI

Machine learning (ML)

Models that learn from data.

06/16

DATA & AI

Data science

Turning raw data into decisions.

07/16

DATA & AI

Data analytics

Dashboards, metrics and insight.

08/16

ENGINEERING

DevOps

Continuous integration. Zero friction.

Cloud computing

AWS / Azure / GCP

Cybersecurity

Blockchain development

Internet of things

Connected edge

Social media management

09/16

ENGINEERING

Cloud computing

AWS / Azure / GCP

10/16

ENGINEERING

Cybersecurity

Defend systems before they are tested.

11/16

ENGINEERING

Blockchain development

Decentralised applications and ledgers.

12/16

ENGINEERING

Internet of things

Connected edge

13/16

GROWTH

Social media management

Voice, community and consistent presence.

Deep learning

14/16

DATA & AI

Deep learning

Layered networks for complex patterns.

Natural language processing (NLP)

15/16

DATA & AI

Natural language processing (NLP)

Software that reads and writes language.

Business analytics

16/16

DATA & AI

Business analytics

Data framed around business questions.

Ready to pick a path?

Read how the BitElevate program works, or talk to the team.

Explore the ecosystem

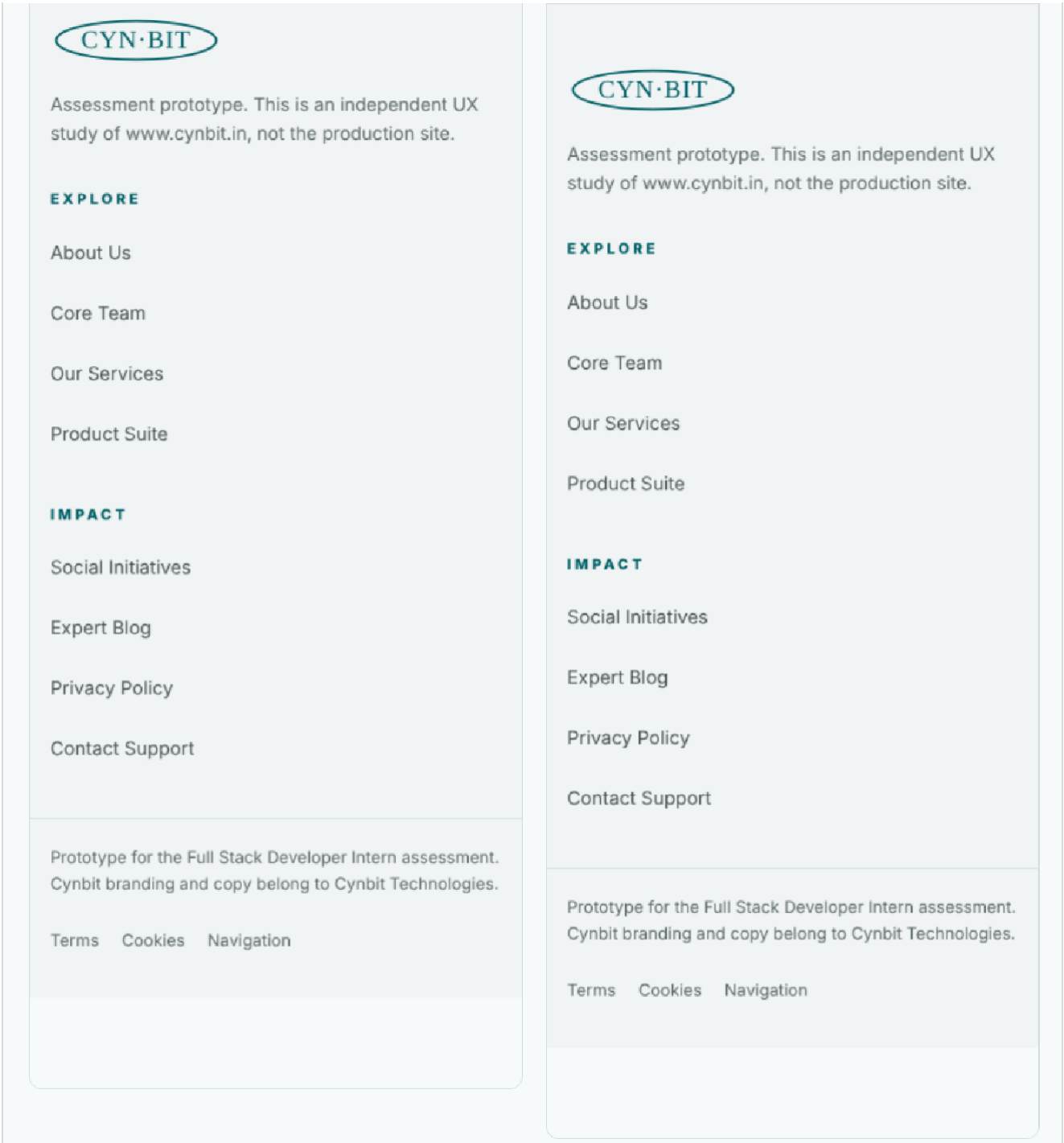
Speak to a mentor

Ready to pick a path?

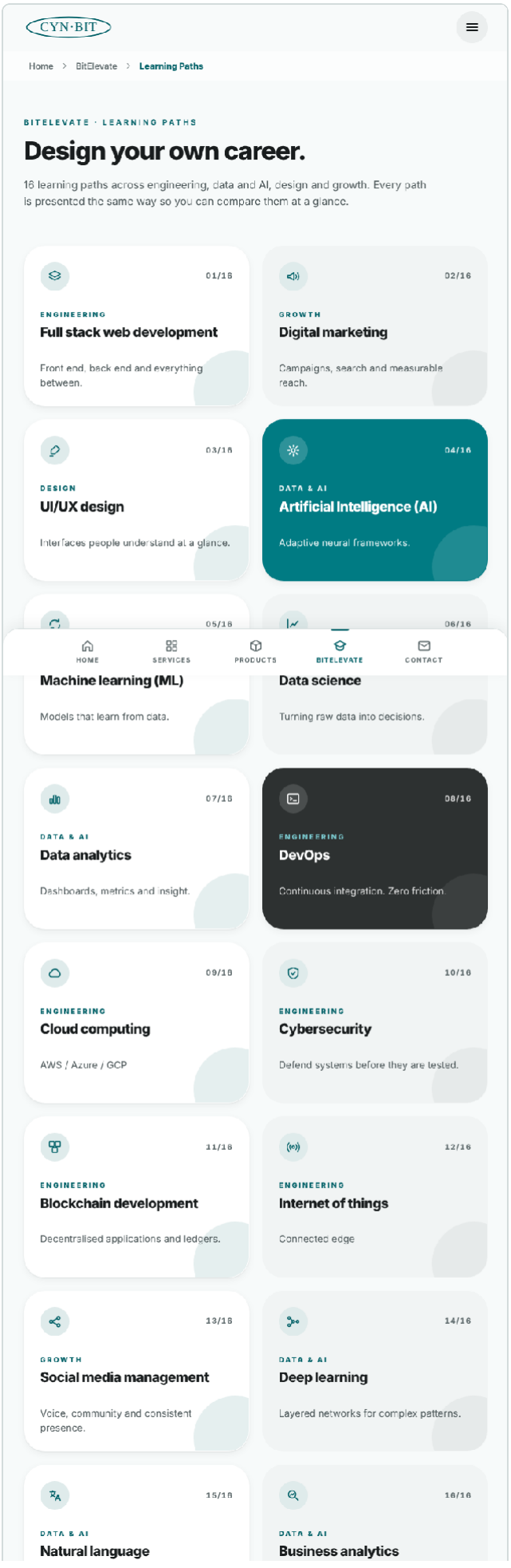
Read how the BitElevate program works, or talk to the team.

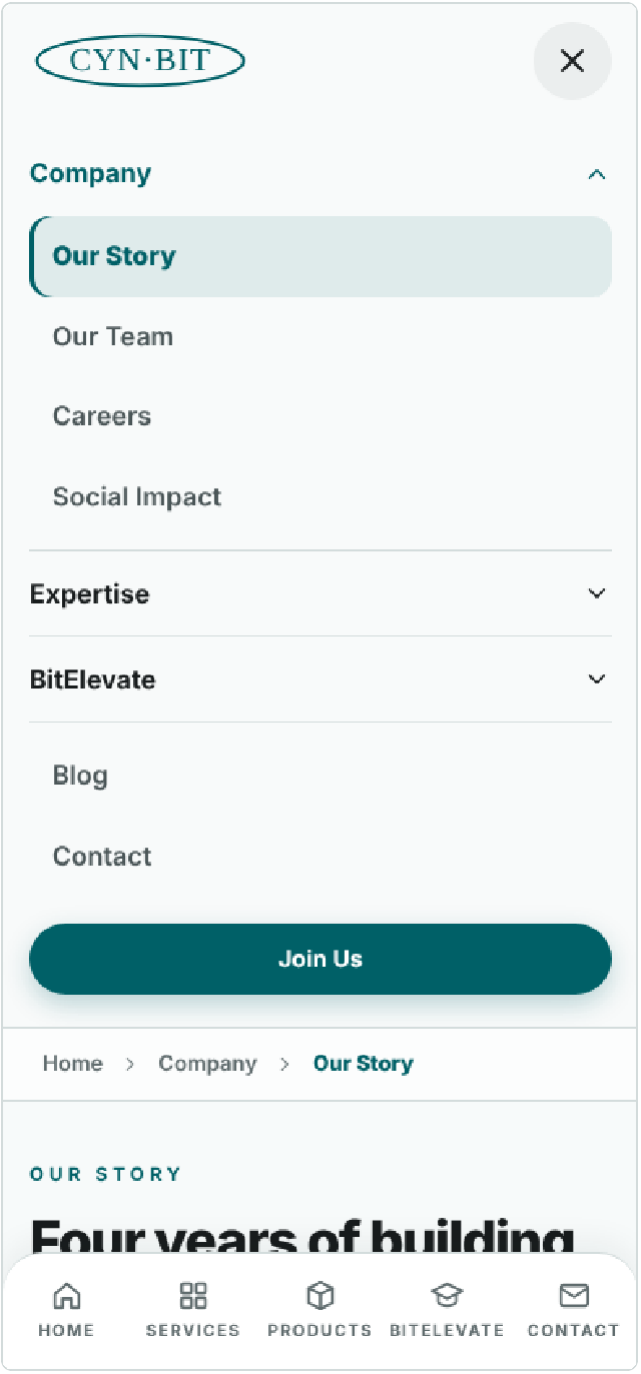
Explore the ecosystem

Speak to a mentor

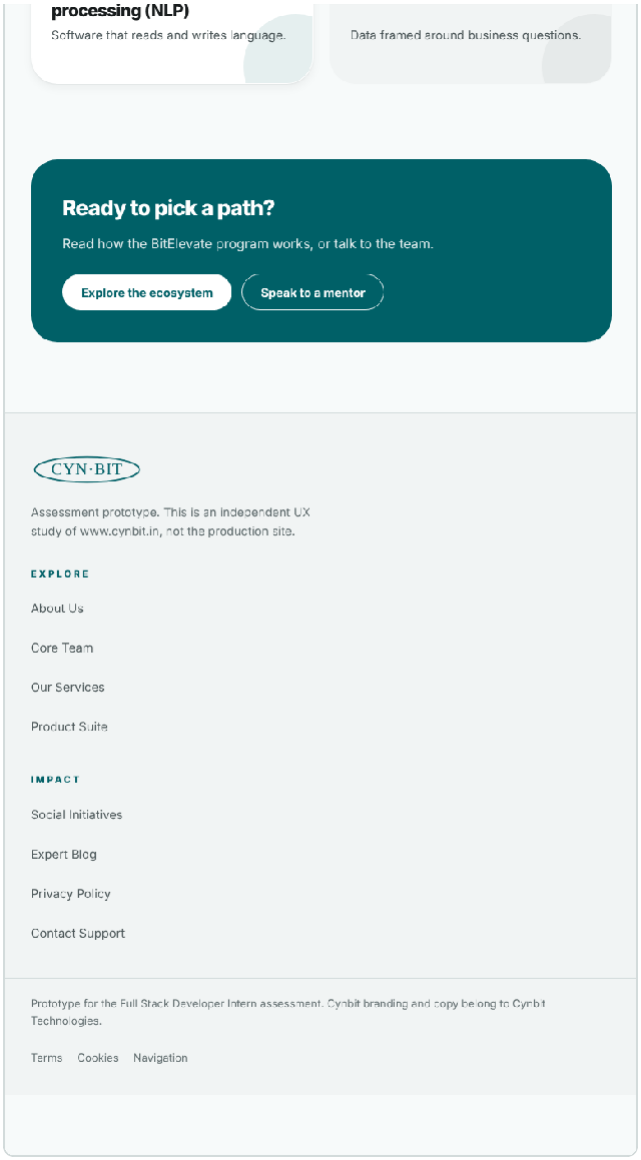


Learning paths, mobile





Improved mobile menu with current page marked



Improved learning paths at 768 px (2 columns)

11. Testing performed and results

An automated Playwright (Chromium) suite, `tests/run-tests.mjs`, was run against the built site. Result: **187 passed, 0 failed**. Full log: `docs/test-results.txt`.

Area	Checks passed	Examples
Routes, console, links	73	All 18 routes return 200; no JS errors; every internal link resolves; one h1 per page; in-page anchors resolve
Active nav + breadcrumbs	26	Correct group, item and trail on 5 routes; header, trigger and breadcrumb stay visible after scrolling; dropdown open, Esc, outside click, select; current item marked on reopen
Learning path cards	26	At 1440, 1024, 768 and 390 px: 16 cards, identical width and height, expected column count, consistent gaps (within 1 px rounding), no horizontal scroll, tags aligned per row

Timeline	7	Year size and contrast measured against the original reconstruction; four milestones with four rail steps; scroll states and spine fill
Mobile navigation	14	Menu toggle, Esc, current item, bottom-bar active state, no overflow on 5 pages
View switch / compare	5	Switch toggles skins; compare frames render original and improved; page selector updates both
Accessibility (Lighthouse items)	30	Headings never skip a level, lists contain only li, touch targets at least 24 px, on 5 pages at 1440 and 390 px
Accessibility basics	6	lang, skip link, labelled landmarks, aria-expanded, SVG labelling

Issues found and fixed during testing: (1) heading levels skipped on Learning Paths (h1 to h3); (2) footer and text links only 18 to 21 px tall; (3) tags misaligned across a card row when descriptions had different lengths (fixed by reserving two lines); (4) the first reconstruction of the original wrongly showed no bottom-bar highlight, corrected after the candidate's screenshots showed Home and Contact highlighted. Several test assertions were also corrected for 1 px sub-pixel rounding, a selector, and a transition delay (test errors, not product defects).

Not done: Lighthouse, screen-reader testing, real-device testing, Safari/Firefox, and any testing against the live site.

12. Limitations, recommendations and future scope

Limitations

- The prototype is not built on Next.js; port the three components and CSS into the real app (mechanical, see README).
- The "original" view is a reconstruction; measurements of the live DOM may differ.
- No Lighthouse run was made on the prototype, and none of the live site's performance problems are claimed fixed.
- Some copy was written for the prototype: 12 card descriptions, category labels, and paraphrased milestone bodies for 2022 to 2024. Replace with Cynbit-approved text.

- Team photos and logo are placeholders (initials, stand-in wordmark).

Recommendations for the live site (each to be verified in the codebase)

1. Add `aria-current` and an active style in the shared Header and bottom nav using `usePathname()`, and a breadcrumb in the layout (the change this prototype demonstrates).
2. Replace the Material Symbols stylesheet with inline SVG icons or a self-hosted subset to remove a render-blocking third-party request (findings 2).
3. Load Google Analytics after idle or interaction (for example Next.js `strategy="lazyOnload"`) and review which components must be client components, then run the Next.js bundle analyzer on the 158 KiB of unused JavaScript (finding 1).
4. Open the Lighthouse accessibility report and fix the listed elements: contrast, list markup, heading order, touch targets (finding 3).
5. Set `og:url` and canonical per page via `generateMetadata`; use a PNG share image of about 1200 < 630 px instead of an SVG logo (finding 5).
6. Consider HSTS, CSP, COOP and frame-protection headers in `next.config` headers (finding 4).

Future scope

Re-run Lighthouse after changes, run an axe audit, test with a screen reader and on real phones, and user-test the breadcrumb on mobile.